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? FR: Configuration Migration Utilities

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v1.0

Feature request

Component	Complexity	Budget (days)

Server Middle 10

Stakeholders sign off status

Role	Name	Sign-off	Version
Lead product owner	@Aleksandrs Kalimulins	?	1.0
Product owners	@Martins Orinskis	?	1.0
	@Dmitrijs Saldovs	👍	1.0
	@Aleksander Vladishev	?	1.0
	@Dmitrijs Jesilevskis	?	1.0
Optional reviewers (none or more)	@Alexei Vladishev	?	1.0
	@Andris Zeila	?	1.0

FR principles

1) focus on user experience first 1.1) in widgets: hide UI elements, instead of disabling them 2) create all UI mockups 3) entity lists must be limited/paginated 4) align with Cloud

Summary

Currently, Zabbix does not provide a built-in way to export and import complete configuration data between installations.

Introduce a standalone command-line utilities, **zabbix_export**, that packages configuration data into a portable format and **zabbix_import**, that restores it into a schema-compatible target database.

Use cases

1. As a Zabbix administrator, I want to export configuration from one Zabbix installation and import it into another clean installation, so that I can quickly replicate, restore or migrate an environment.
2. As a Zabbix administrator, I want to migrate configuration from a PostgreSQL-based Zabbix installation to a clean MySQL-based installation, so that I can change the database engine without manually recreating configuration

Zabbix acceptance

1. Create standalone command-line **zabbix_export**, **zabbix_import** utilities
 - a. Operate independently of Zabbix Server, Frontend, and Proxy runtime processes
 - b. Use a database-independent package format for configuration export and import
 - c. Provide machine-readable output for automation and external integrations
2. **zabbix_export**:
 - a. Parameters:
 - `--db-type mysql|postgresql`
 - `--db-host host`
 - `--db-port port`
 - `--db-name name`
 - `--db-user user`
 - `--db-password password`
 - `--output file`Default: **zabbix-config-YYYYMMDD-HHMMSS.tar.gz**
 - b. On execution:
 - i. connects to the source database
 - ii. exports all database tables to CSV files (Appendix A) except those listed in Appendix B
 - iii. generates manifest.json containing package metadata and table statistics
 - iv. generates checksums.sha256 containing a SHA-256 checksum for each exported file
 - v. packages all files into archived file (tar + gzip) following the layout defined in Appendix C
3. **zabbix_import**:
 - a. Parameters:
 - `--input file`
 - `--db-type mysql|postgresql`
 - `--db-host host`
 - `--db-port port`
 - `--db-name name`
 - `--db-user user`
 - `--db-password password`
 - b. Successful import requires a target database schema compatible with the exported package
 - c. On execution:

- i. verifies bundle integrity (checksums, manifest, row counts) before touching the target database
- ii. disables foreign key checks
- iii. loads CSV data
- iv. validates post-load row counts for each imported table
- v. recreates excluded "ids" table entries
- vi. enables foreign key checks

Appendix A — CSV file format

1. CSV format follows RFC 4180 quoting and field formatting rules:
 - a. Encoding: UTF-8
 - b. Separator: comma (,)
 - c. First row contains quoted column names
 - d. Every non-NULL field enclosed in double quotes ("")
 - e. Embedded double quotes escaped by doubling (""")
 - f. Backslash (\) has no special meaning and treated as literal data
 - g. SQL NULL values are represented by an unquoted NULL token
 - h. The quoted value "NULL" must be interpreted as a literal string value and not as SQL NULL.
 - i. Embedded CR, LF, or CRLF characters are permitted inside quoted fields

Appendix B — Default excluded tables

acknowledges, alerts, auditlog, autoreg_host, changelog, dbversion, dhosts, dservices, escalations, event_recovery, event_suppress, event_symptom, event_tag, events, ha_node, history, history_bin, history_json, history_log, history_str, history_text, history_uint, host_rtdata, housekeeper, ids, item_rtdata, lld_macro_export, problem, problem_tag, proxy_autoreg_host, proxy_dhistory, proxy_group_rtdata, proxy_history, proxy_rtdata, service_alarms, service_problem, service_problem_tag, sessions, task, task_acknowledge, task_check_now, task_close_problem, task_data, task_remote_command, task_remote_command_result, task_result, trends, trends_uint, trigger_queue

Appendix C — Bundle format

```
zabbix-config-YYYYMMDD-HHMMSS.tar.gz
```

```
|— manifest.json
|— checksums.sha256
|— users.csv
|— hosts.csv
|— items.csv
|— ...
```

Archive

- Format: tar + gzip (.tar.gz produced as zabbix-config-YYYYMMDD-HHMMSS.tar.gz)
- Maximum compressed size: **5 GB** (S3 single-PUT limit)

manifest.json

Defines general attributes and basic metadata per table. Schema:

```
{
  "format_version": 1,
  "zabbix_version": "7.0.5",
  "dbversion_mandatory": 7000000,
  "created_at": "2026-07-01T10:00:00Z",
  "tables": [
    { "name": "role", "rows": 4 },
    { "name": "hosts", "rows": 312 }
    ...
  ]
}
```

checksums.sha256

- One line per file in the bundle, covering metadata file and all CSV files
- Format: standard sha256sum output — <hex_digest> <relative_path> (two spaces between digest and path), one entry per line

Example:

```
3f0f4781f68f0e1bcb7f0f6c8d0f54f6d4f2d7a1d8c8b9e1a4b8c3d2e1f0a123  manifest.json
e1a2b3c4d5e6f7890123456789abcdef0123456789abcdef0123456789abcdef  hosts.csv
9f8e7d6c5b4a32100123456789abcdef0123456789abcdef0123456789abcdef  items.csv
```

Zabbix UI changes

N/A

Open questions

N/A

Decisions made

1. Decision

Change log

N/A



0 Likes

[\(/xwiki/bin/view/Team%20Spaces/DEV/Design%20specifications/Zabbix%208.2/Requirements%208.2/%E2%9C%8E%20FR%3A%20Configuration%20Migration%20Tool/?viewer=likers\)](/xwiki/bin/view/Team%20Spaces/DEV/Design%20specifications/Zabbix%208.2/Requirements%208.2/%E2%9C%8E%20FR%3A%20Configuration%20Migration%20Tool/?viewer=likers)

Tags:

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